



Time: 5 minutes

Outcome: Determine equilibrium, shortages, surpluses, adjustment, and comparative statics.

Difficulty: Beginner

Shortages and Surpluses

THE CORE IDEA

A shortage occurs when quantity demanded exceeds quantity supplied at the current price. Buyers competing for limited supply put upward pressure on price. A surplus occurs when quantity supplied exceeds quantity demanded; sellers then face pressure to lower price. Adjustment tends toward the price where the two quantities are equal.

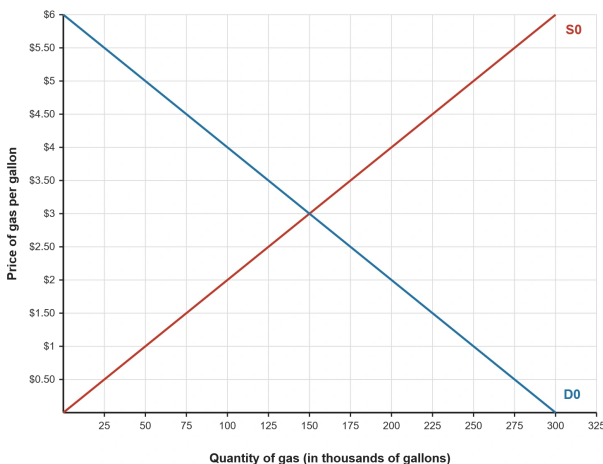
HOW TO RECOGNIZE IT

- At the current price, buyers want 180 units and sellers offer 120 units.
- A price above equilibrium creates a surplus because quantity supplied exceeds quantity demanded.
- A price below equilibrium creates a shortage because quantity demanded exceeds quantity supplied.
- A surplus exists when quantity supplied is greater than quantity demanded.

WATCH OUT

A common mistake is to reverse shortage and price pressure. Quantity demanded exceeds quantity supplied, so there is a shortage. Shortages push price upward.

WORKED EXAMPLE: MEASURING A SHORTAGE



At a price of \$2 per gallon, the graph shows quantity demanded of 200 thousand gallons and quantity supplied of 100 thousand. Because buyers want 100 thousand more gallons than sellers offer, the market has a shortage of $200 - 100 = 100$ thousand gallons. Upward pressure on price follows.

CHECK YOURSELF

At a price of \$4, buyers want 100 thousand gallons and sellers offer 200 thousand. Calculate the imbalance and predict the pressure on price.